
MP Online • Internship Project • 2025-26

GrievanceHub

Digital Grievance Redressal System

Process Compliance Document

Standards, Best Practices & Methodology Followed

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MP Online Internship Project

1. Overview

This document describes the development standards, coding practices, and software engineering principles followed during the development of GrievanceHub. It demonstrates adherence to professional software development processes appropriate for an academic/internship-level project.

2. Software Development Methodology

Aspect	Approach Followed
Development Model	Iterative and Incremental - features built and tested phase by phase
Architecture Pattern	Layered Architecture (Models > Services > Routers) on backend; Component-based on frontend
API Style	RESTful API design with proper HTTP methods (GET, POST, PATCH) and status codes
Version Control	Git-based source control with organized folder structure
Documentation	Swagger/OpenAPI auto-docs via FastAPI; separate project docs prepared

3. Coding Standards

3.1 Backend (Python / FastAPI)

- PEP-8 naming conventions: snake_case for functions and variables
- Separation of concerns: models, schemas, services, and routers kept separate
- Dependency injection used for DB sessions and auth guards (FastAPI Depends)
- Pydantic v2 schemas used for all request/response validation
- Async/await used for I/O-bound operations in grievance submission
- Sensitive config (DB credentials, secret keys) stored in .env file, not in code

3.2 Frontend (React / JavaScript)

- Functional components used throughout - no class components
- React hooks (useState, useEffect, useContext) used for state management
- Custom hooks created for reusable logic (useAuth, useGrievance)
- Axios instance configured centrally with base URL and interceptors
- Context API used for global auth state management
- Components organized by feature: common/, grievance/, admin/

4. Security Compliance

Requirement	Implementation	Status
Password Security	Bcrypt hashing via Passlib - plaintext passwords never stored	Done
Authentication	JWT tokens with 60-minute expiry via python-jose	Done
Authorization	Role-based guards on all protected routes	Done
Input Validation	Pydantic v2 validates all API input automatically	Done
CORS Policy	Restricted to http://localhost:5173 only	Done
Secret Management	.env file used; .gitignore excludes it from version control	Done

5. Documentation Compliance

Document	Status
Development Planning Document	Completed
Business Requirement Document (BRD)	Completed
Design Document (DDD)	Completed
Requirements Traceability Matrix (RTM)	Completed
Process Compliance Document	Completed
Test Document	Completed
User Guide	Completed